

Physical Data Model

After discussing and analyzing, we choose :

- DBMS: SQLite
- Data Model Type: Relational

1. Order Table

#	PK	FK	Column Name	Data type	Mandatory	Description
1	x		orderID	INTEGER	Yes	Unique identifier for the order
2			status	VARCHAR	Yes	Current status: "pending," "approved," "rejected," "canceled".
3			shippingFee	FLOAT	Yes	Calculated VAT amount (10% of subtotal)
4		x	deliveryInfoID	INTEGER	Yes	Reference to delivery information (FK to DeliveryInfo)

2. OrderItem Table

#	PK	FK	Column Name	Data type	Mandatory	Description
1	x	x	orderID	INTEGER	Yes	Reference to the order (FK to Order)
2	x	x	productID	INTEGER	Yes	ID of the product in the order
3			quantity	INTEGER	Yes	Quantity of the product ordered
4			unitPrice	FLOAT	Yes	Price of the product at time of order
5			subtotal	FLOAT	Yes	Total price for this item (unitPrice × quantity)
6			isRushDeliveryEligible	BOOLEAN	Yes	Whether this item is eligible for rush delivery
7			weight	FLOAT	No	Weight of one unit of this product

3. Cart Table

#	PK	FK	Column Name	Data type	Mandatory	Description
1	x		cartID	INTEGER	Yes	Unique identifier for the cart
2			sessionID	VARCHAR	Yes	Identifier for the user's session
3			createdDate	DATE	Yes	Date when the cart was created
4			updatedAt	DATE	Yes	Date when the cart was last updated

4. CartItem Table

#	PK	FK	Column Name	Data type	Mandatory	Description
---	----	----	-------------	-----------	-----------	-------------

1	X	X	cartID	INTEGER	Yes	ID of the cart containing this item
2	X	X	productID	INTEGER	Yes	ID of the product in the cart
3			quantity	INTEGER	Yes	Quantity of the product
4			price	FLOAT	Yes	Selling price of the product

5. Card Table

#	PK	FK	Column Name	Data type	Mandatory	Description
1	x		cardID	INTEGER	Yes	Unique identifier for the card
2			cardCode	VARCHAR	Yes	Credit card number
3			owner	VARCHAR	Yes	Name of the cardholder
4			cvvCode	VARCHAR	Yes	Card verification value/code
5			dataExpired	DATE	Yes	Expiration date of the card